Ahmed Abdullah

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Education

- **National University of Computer and Emerging Sciences** — Lahore, Pakistan
  Bachelor of Science in Data Science; GPA 3.6 — *Aug. 2022 – May. 2026*
  - **Honors**: Dean's List, Top of Batch
  - **Experience**: Teaching Assistant for OOP (CS1004) and COAL (EE2003)
  - **Relevant Courses**: Data Structures, Adv. Statistics, Artificial Intelligence, Operating Systems, Data Mining, Parallel and Distributed Computing & Data Warehousing.

Experience

- **College of Engineering and Computing - Missouri S&T University** — Missouri, US (Remote)
  Research Assistant — *Feb 2025 – Present*
  - **Computational Biology Research**: Leading a team of 3 junior engineers to build a predictive model for tumor reduction using machine learning. Leveraged large-scale patient data from 50,000+ patients to assess the efficacy of cancer drugs over time. Contributed to designing the modeling pipeline and ensuring accurate insights for cancer growth.
- **MacroMed (Pvt.) Ltd.** — Lahore, PK (Hybrid)
  AI Engineer Intern — *Mar 2025 – May 2025*
  - **Multimodal AI Infrastructure**: Led a team of 3 AI engineers to develop a Level 2 API for multimodal product recommendations, order management, and dynamic FAQs. Integrated LangChain, Selenium Automation, and RAG-enabled LLMs for real-time internet search. Reduced inference time to 500ms while enabling image-text understanding and retrieval using ImageBind.
- **Multimedia Mining & Search Group - Johannes Kepler University** — Linz, AT (Remote)
  Research Assistant — *Aug 2024 - Feb 2025*
  - **Face-Voice Association with Missing Modalities**: Developed a deep learning system for associating faces with voices, boosting model accuracy by 12% on a widely used benchmark (VoxCeleb). Leveraged advanced audio-visual models like WavLM, ECAPA, and VGGVox to achieve state-of-the-art performance. Redesigned the training pipeline, introducing reusable modules that cut processing time by 10% and reduced setup time by 70%, significantly improving development efficiency.
- **Machine Intelligence Group - FAST-NUCES** — Lahore, PK
  Research Intern — *Aug 2024 - Nov 2024*
  - **Predicting Outcome for Ischemic Stroke:**: Designed an AI system to predict stroke outcomes using MRI scans. Led the effort to improve model performance by enhance image data and increasing accuracy from 44% to 57%. Contributed to optimizing the image augmentation process, helping to integrate different data sources and set a new benchmark for stroke prediction.

Projects

- **CrickAI – LBW Review System using YOLO**: Built a video review tool for LBW decisions in cricket using YOLO for object detection. Utilized Roboflow for dataset preparation and LabelImg for annotation.
- **PolyHope-M – Hope Speech Classification**: Built a multilingual classification model using LLaMA and PyTorch to classify supportive speech online. Achieved 1st place out of 400+ submissions on CodaLab with top-ranked accuracy.
- **DermaScan3D – Skin Cancer Detection using CNNs**: Built a multimodal TensorFlow model with ResNet for skin cancer classification. Ranked in the top 1% (8000+ participants) on Kaggle.
- **VisionFind – Multimodal Search Engine using CLIP**: Built a semantic search engine using CLIP, HuggingFace, and PyTorch to retrieve images and text based on visual and textual queries.

Programming Skills

- **Languages**: Python, C++, C, JavaScript.
- **Technologies**: PyTorch, TensorFlow, Keras, HuggingFace, MONAI, Scikit-learn, MLLib, Transformers, OpenCV.
- **Tools & Platforms**: LangChain, Flask, Firebase, Github, PowerBI, Tableau, Hadoop, PySpark, PostgreSQL, MongoDB.